

REQUIREMENT ANALYSIS PHASE

4. Customer Journey Map

Customer Journey for Retail Manager (Rajesh)

Stage	User Actions	User Thoughts	Pain Points	Opportunities
Awareness	Views weekly sales reports	"Sales are inconsistent"	Manual report generation	Automated dashboards
Consideration	Explores data patterns	"Where are the best placements?"	Data silos across systems	Integrated data views
Decision	Selects product categories to analyze	"This category needs attention"	Unclear cause-effect relationships	Interactive filtering
Purchase	Implements placement changes	"Hope this improves sales"	Execution challenges	Real-time updates
Retention	Monitors performance	"Is this working?"	Tracking effectiveness	Performance metrics

5. Solution Requirements

(4 Marks)

Functional Requirements:

FR No.	Epic	Sub Requirement (Story / Sub-Task)
FR-1	Data Import & Management	Import CSV dataset from Kaggle; Validate types; Handle missing values; Create Calculated Fields (Price Difference, Sales per Price, etc.)
FR-2	Data Visualization	Create Bar (Sales per Category), Stacked Bar (Promotion), Donut (Position), Scatter (Competitor Price), Line (Seasonal), Pie (Demographics), Bubble (Traffic), Heat Map.
FR-3	Dashboard Development	Combine 8 visuals; Add layout containers; Implement interactive filters; Enable drill-down.
FR-4	Story Development	Create Tableau Story; Add captions and annotations.
FR-5	Web Integration	Build Bootstrap website; Embed Tableau Dashboard and Story.
FR-6	Publishing	Publish to Tableau Public; Share public URL.

Non-functional Requirements:

NFR	Requirement	Description
NFR-1	Usability	User-friendly; responsive design; intuitive filters.
NFR-2	Performance	Dash load < 3s; instantaneous filter updates.
NFR-3	Accessibility	WCAG compliance; contrast ratios; keyboard support.
NFR-4	Scalability	Datasets up to 50,000+ records.
NFR-5	Compatibility	Cross-browser (Chrome, Firefox, Safari, Edge).
NFR-6	Maintainability	Documented workbooks; clean layout; clean CSS.

6. Data Flow Diagram

(4 Marks)

Step	Process	Description
------	---------	-------------

1	Data Collection	CSV from Kaggle (~1,000 product records)
2	Data Import	CSV to Tableau Desktop
3	Data Validation	Confirm 1,000 records loaded
4	Data Cleaning	Check missing/duplicates
5	Calculated Fields	Price Difference, Promo Status, Discount Category, Sales per Price
6	Viz Creation	Generate all 8 visualization types
7	Dashboard	Combine with filters
8	Story	Narrative with captions
9	Publishing	Upload to Tableau Public
10	Web Integration	Embed in Bootstrap HTML

7. Technology Stack

(4 Marks)

#	Description	Technology
1	Dataset Source	Kaggle
2	Data Processing	Tableau Desktop
3	Data Visualization	Tableau Desktop, Tableau Public
4	Dashboard	Tableau Desktop (Containers, Filters, Actions)
5	Story	Tableau Story (Captions, Annotations)
6	Web Framework	HTML5, CSS3, Bootstrap 5